AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT132 – M132
Linear Algebra

Final Examination
Fall Semester 2025/2026

Date:14 /January/2026

Number of Exam Pages: 3 pages (including this cover sheet)	**Time Allowed:** 2 Hours

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Q–1: [5×3 marks] Choose the correct answer:

1.1: The invers of $\begin{bmatrix} 2 & -2 \\ 2 & 2 \end{bmatrix}$

a) Not invertible since its determinant is 0
b) $\begin{bmatrix} 0.25 & 0.25 \\ 0.25 & 0.25 \end{bmatrix}$
c) $\begin{bmatrix} 0.25 & 0.25 \\ -0.25 & 0.25 \end{bmatrix}$
d) $\begin{bmatrix} 0.25 & -0.25 \\ 0.25 & 0.25 \end{bmatrix}$
e) None of the above

1.2: If $W = Span\left\{\begin{bmatrix} 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 7 \\ 2 \end{bmatrix}\right\}$, then $\dim(W) =$

a) 0
b) 1
c) 2
d) 3
e) None of the above

1.3: The linear system $AX = B$, $A = \begin{bmatrix} 2 & -4 \\ a^2 & -8 \end{bmatrix}$ does not have a unique solution at

a) $a = 0, 4$
b) $a = 2, -2$
c) $a \neq 0, 4$
d) $a \neq 2, -2$
e) None of the above

1.4: The vector that can be written as a linear combination of $v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, and\ v_2 = \begin{bmatrix} 4 \\ -3 \end{bmatrix}$ is

a) $u = \begin{bmatrix} 6 \\ 7 \end{bmatrix}$
b) $u = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$
c) $u = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$
d) $u = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$
e) None of the above

1.5: If $A = \begin{bmatrix} 2 & 0 & -3 \\ 4 & 1 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -1 & -2 \\ 3 & 4 & -7 \end{bmatrix}$, then

a) $a_{12} + b_{12} = 1$
b) $a_{12} - b_{12} = -1$
c) $a_{21} + b_{21} = 7$
d) $a_{21} - b_{21} = -1$
e) None of the above

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Q–2: [8 marks] Answer the following questions as True or False (2 mark), Justifying your answer (2 marks):

2.1: The linear system $\left[\begin{array}{cc|c} 1 & -2 & 2 \\ a & 2 & 6 \end{array}\right]$ has a unique solution only at $a = -1$.

2.2: For a square matrix A, if A is invertible then A^T is also invertible.

Each question is worth 9 marks. Answer the following questions:

Q–3: [5+2+2 marks]

Let $W = \left\{ \begin{bmatrix} a+b \\ 2a-3b \\ c \end{bmatrix} : c = a-b, a, b \in R \right\}$ be a subset of R^3.

a) Show that W is a subspace of R^3.
b) Find a basis for W.
c) Is $\begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} \in W$. Explain.

Q–4: [3+4+2 marks]

Define the mapping $T: R^3 \to R^2$ by $T\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2x+y \\ 2y+z \end{bmatrix}$.

a) Show that T is a linear transformation.
b) Find the null of T, $N(T)$ and range of T, $R(T)$.
c) Find $Dim(N(T))$ and $Dim(R(T))$.

Q–5: [9 marks]

Find the eigenvalues of the matrix $A = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$; then find matrices P and D such that $A = PDP^{-1}$.

End of questions

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